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IoT-driven smart agricultural technology for real-time soil and crop optimization

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ABSTRACT

To address world food and agricultural sustainability, the UN's Sustainable Development Goals (SDGs) place strong emphasis on a transformative agricultural sector. By 2050, there will be about 10 billion people on the planet, making it imperative to increase about 60 percent in food production efficiency and security in order to meet the growing population's demands. In this critical situation, IoT-based smart farming offers a promising solution that can achieve 12 SDGs out of 17. This study unveils a groundbreaking Internet of Things (IoT) system designed to revolutionize agricultural practices by providing real-time monitoring of soil parameters, including temperature, moisture, salinity, EC, pH, nitrogen, potassium, and phosphorus levels. Field testing conducted with rice crops in Rahim Yar Khan, Pakistan, demonstrated the system's remarkable ability to accurately measure all eight critical parameters. The soil analysis showed a temperature between 30.5 °C and 33.2 °C, a moisture content between 60.6 % and 94.1 % and a pH value between 7.13 and 8.33. Nutrient content varied with nitrogen (71–103 mg/kg), phosphorus (15–19 mg/kg), and potassium (101–141 mg/kg). Leveraging this data, an AI-driven mobile application was used to deliver best recommendations for optimizing crop management, particularly in fertilization, irrigation and disease diagnoses practices. By integrating advanced IoT technologies, cloud computing, predictive algorithms, and a smart soil sensor, this system revolutionizes agriculture by enabling real-time monitoring of critical factors influencing rice crops metabolism. It empowers farmers with data-driven insights to enhance productivity, optimize resource use, and improve sustainability. This scalable solution addresses modern agricultural challenges by reducing environmental impact and strengthening crop resilience. Focused on unlocking the potential of data-driven decision-making, this research paves the way for sustainable, smart farming practices to meet the growing demands of a rapidly increasing global population.

1. Introduction

With a growing global population comes the rising issue of food security. One of the solutions, which will definitely carry the day, is optimizing agricultural productivity by smarter resource management. However, four significant challenges hold back the efficient and sustainable development of agriculture. The first one is that labor-intensive and time-consuming manual labor severely curbs productivity. The second is poor decision-making; decision-making frequently quite relies on guesswork to impede the attainment of optimal crop management. Specific concerns are also complicated by issues such as climate change, whose impacts include irregular weather patterns and degraded soil quality. Finally, inefficient use of resources, including overuse of water and fertilizers, not only raises costs but also adversely affects the environment [1,2].

Agriculture-based industries act as backbones of economies in developing countries, such as Pakistan [3]. Healthy soils, crop growth analysis and optimization of crop metabolism determine sustainable crop production and food security in developing countries. Productivity increases due to improved farming practices for soil management and crop health monitoring, while conserving environmental resources for future generations [4]. Agriculture is a cornerstone of Pakistan's economy and a vital occupation for human survival. Its success hinges on healthy soil and farmers' knowledge of crop and land management.

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Healthy soil also supports sustainable crop production by acting as a reservoir for nutrients and moisture, essential for plant growth and microbial activity. Informed farmers can make better decisions regarding planting, irrigation, and fertilization, ensuring optimal productivity [5]. These factors collectively sustain agricultural output and contribute to the country's economic stability [6,7]. It also makes a significant contribution to our daily lives [8]. For farmers, understanding soil nutrients is critical to success [9–11].

The literature highlights various IoT-driven farming systems for soil monitoring and crop management. In study of Zhang et al., 2017 IoT-based systems with sensors offer high accuracy for parameters like soil temperature and nutrients but lack mobile application integration [12]. Abishek et al. (2021) analyzed smart irrigation systems and IoT kits that mainly monitor soil moisture and weather conditions. Their research found moderate accuracy and cost-effectiveness, but also found that there is no cloud dashboard integration [13]. In the study by Mesas-Carrascosa et al. (2015), IoT-based irrigation systems were examined for their efficiency in monitoring soil and climate parameters. The study found that while mobile app support was available, cloud dashboard functionality was not included [14]. In the research of Ayaz et al. (2019) and Pitakaso et al. (2023), Decision Support Systems and AMIS optimize farming decisions but miss mobile support [15,16]. The extensive research and development and literature on IoT-based nutrient management also highlights innovative approaches namely precision farming—spanning automated devices [17], drones [18], satellite imagery [19], big data [20], and machine learning [21] to improve agricultural efficiency and sustainability. A research study by Hartono et al. (2024) presents a portable IoT system for real-time monitoring of soil nutrients, focusing on nitrogen, phosphorus, and potassium, which helps farmers make informed fertilizer decisions [22]. In the study of Islam et al. (2023), an advanced machine learning IoT-based device has been developed to monitor soil conditions and provide custom-made crop recommendations, optimizing resource use and increasing productivity [23]. Furthermore, the integration of IoT and machine learning into nutrient management models enables effective yield prediction and nutrient loss analysis, addressing the challenges of environmental factors [24]. While numerous studies highlight the transformative potential of IoT in nutrient management, the lack of comprehensive insights into crop health and key parameters of soil leaves a critical research gap. Soil key parameters must be monitored: temperature, moisture, salinity, pH, and nitrogen (N) to allow for the best environment for plant growth, nutrient uptake, and soil condition. All these make up the foundation of farming. Nevertheless, by also measuring electrical conductivity (EC), potassium (K), and phosphorus (P), farmers can improve their agricultural productivity management to a large extent. The inclusion of these extra parameters makes it possible to obtain a better understanding of the state of the soil, and therefore, healthier crops, higher yields, and, again, higher yields.

To address these challenges, it is essential to monitor not only soil nutrition but also crop growth and metabolism. Knowledge as to how plants metabolize nutrients as well as the impact of soil nutrition on their growth permits a farmer to make an informed judgment regarding fertilizer, irrigation practices, and overall crop management. Monitoring key soil parameters that include nitrogen, phosphorus, and potassium counterpart environmental parameters such as temperature, moisture, and pH levels increase the healthfulness of crop metabolism. Proper nutrient management skills ensure that crops attain all the elements necessary for photosynthesis, energy production, and protein synthesis, as crop health and yield are directly impacted. The application of latest technologies in the context of soil science in respect of sensors technology, IoT-based systems, machine learning, data analytics and cloud computing enable farmers to gain real-time insights into both soil conditions as well as plant metabolism through optimize resource use efficiency.

While IoT offers significant benefits in smart farming, challenges like real-time soil nutrient monitoring and accurate fertilizer and irrigation planning remain critical. The overall goal of this research is to show how the use of a smart system that measures eight key parameters of soil and the use of an AI-driven mobile app can change the world of eco-agriculture and increase food security. This study introduces a cutting-edge IoT system for real-time monitoring of soil parameters, including temperature, moisture, and nutrients, enhancing rice crop management. Field tests in Rahim Yar Khan, Pakistan, showcased its accuracy in measuring critical factors affecting crop metabolism. Supported by an AI-driven mobile app, it empowers farmers with data-driven insights to optimize productivity, sustainability, and resource efficiency, addressing modern agricultural challenges. The main objective of the study is to evaluate the effectiveness and accuracy of the proposed IoT-based real-time soil monitoring and data collection system for agricultural applications.

1.1. Key concepts in smart farming

Understanding the basic ideas behind soil, nutrients, and crop health is crucial to comprehending smart agriculture and its uses. An overview of these important soil and crop concepts is provided in this section.

1.1.1. Important parameters for rice crop

Rice cultivation requires specific environmental conditions to ensure optimal growth and yield [25]. Among these, soil temperature, moisture, and pH are critical factors influencing the various stages of rice growth, from germination to ripening. Maintaining appropriate levels of these parameters promotes nutrient availability, root development, and plant health, ensuring a sustainable and productive crop cycle. In Table 1 important parameters of rice crop soil are presented with the summary:

1.1.2. Soil nutrients of rice crop for sustainable production

Soil nutrients are vital for sustainable crop production, influencing plant growth, health, and yield [29]. Key nutrients like nitrogen, phosphorus, and potassium play essential roles in metabolism, structure, and stress resistance. Monitoring and managing these nutrients effectively can optimize crop productivity while ensuring environmental sustainability [30,31]. The following Fig. 1 illustrates various diseases due to nutrient deficiencies in plants. Nutrient deficiencies specially of nitrogen (N), phosphorus (P), and potassium (K) in rice crops can lead to various issues, impacting overall growth and yield. Nitrogen deficiency causes yellowing of older leaves (chlorosis), reduced tillering, and stunted growth, leading to poor grain production. Phosphorus deficiency results in dark green or purplish leaves, delayed maturity, and underdeveloped roots and grains. Potassium deficiency manifests as yellowing or browning of leaf edges, weak stems, and poor grain filling, leaving the crop more vulnerable to diseases and lodging. Addressing these deficiencies is essential to ensure healthy rice crop development and maximize productivity.

1.1.3. The importance of observing plant metabolism in rice cultivation

The metabolism of rice crops is shaped by factors like nutrient levels, water availability, and environmental conditions [32,33]. By using a sensor that tracks key elements such as temperature, moisture, salinity, pH, electrical conductivity, nitrogen, potassium, and phosphorus, farmers can gain real-time insights into the soil condition and crop health. This data helps ensure the rice plants are getting the right nutrients, moisture, and conditions needed for optimal growth. When integrated into a smart farming system with an online cloud and mobile application, the sensor allows farmers to make more informed decisions about when and how to irrigate, fertilize, and manage pests, ultimately boosting rice productivity and promoting more sustainable farming practices. The following Table 2 summarize the crucial constraints for growth of rice crop.

Table 1
Essential parameters of rice crop soil.

Crop Growth Phase	Growth Stage	Soil Temperature (°C)	Soil Moisture Level	Soil pH	Description	References
Establishment	Seed Germination	20–35	70–80 % field capacity	5.5–6.5	Optimal for rapid and uniform germination; excess moisture or improper pH reduces germination rates.	[25]
Vegetative Phase	Tillering	25–30	80–90 % field capacity	5.5–6.5	Ideal for root development and tiller formation; consistent moisture supports nutrient uptake.	[25,26]
Reproductive Phase	Panicle Initiation	22–30	70–80 % field capacity	5.5–6.5	Crucial for panicle development; temperature extremes may impair flowering and grain setting.	[27]
	Flowering	22–30	70–80 % field capacity	5.5–6.5	Stable moisture and temperature required for pollination and fertilization; sensitive to water stress.	[27]
Ripening Phase	Grain Filling	20–27	60–70 % field capacity	5.5–6.5	Cooler temperatures aid uniform grain filling; too much water may delay ripening.	[28]
	Maturation	20–27	50–60 % field capacity	5.5–6.5	Supports grain drying and harvest readiness; low soil moisture is preferred.	[28]

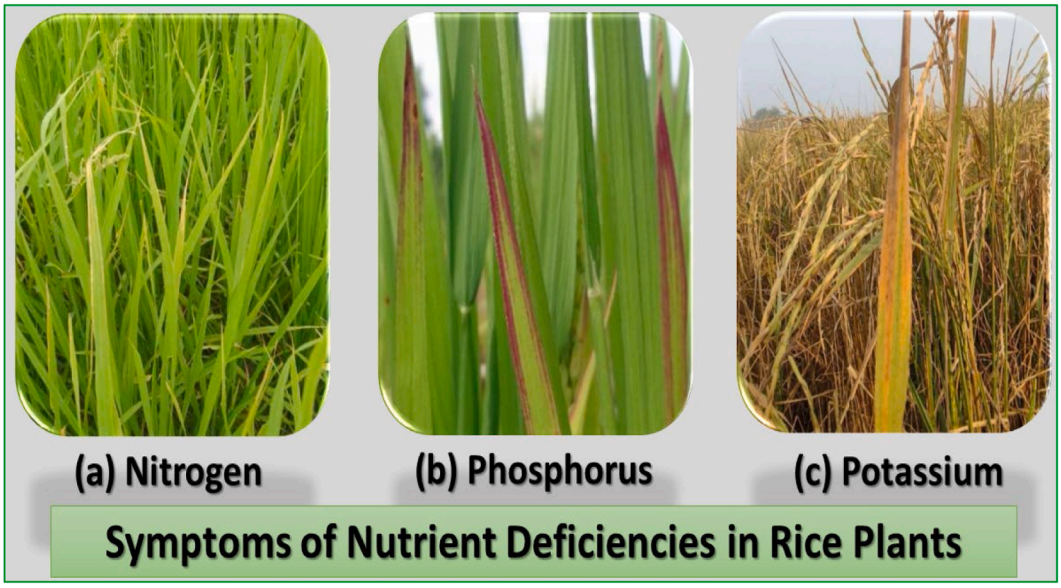


Fig. 1. Nutrient deficiencies in rice crop.

Table 2
Crucial constraints for growth of rice crop.

Parameter	The key feature in management healthy soil	Reference
Phosphorus	Enhances root development, energy transfer, and early growth in rice	[34]
pH	Optimal pH ensures nutrient availability and root health in rice fields	[35]
Moisture	Essential for maintaining flooded conditions and nutrient absorption	[36]
Temperature	The specific level is crucial for germination, vegetative growth, and grain filling in rice crops	[37–40]
Nitrogen	Promotes tillering, leaf growth, and chlorophyll production in rice	[30, 41–44]
Potassium	Improves disease resistance, water regulation, and grain quality in rice	[30]

1.1.4. Internet of Things (IoT) and fertilizer requirement for rice crop growth

Internet of Things (IoT), a modern smart technology, describes a network of smart devices that are linked to the Internet and have the ability to collect, share, and analyse data. These devices often have sensors, software, and other technologies that enable them to communicate with main control systems and other devices, facilitating the automation and remote monitoring of many processes via Internet clouds [45–47]. IoT devices used in agriculture can include soil moisture sensors, weather stations, and NPK sensors, which measure the soil's nutrient content [48,49]. They improve farm management by providing farmers with the ability to obtain real-time information about their farms. For example, as shown in Fig. 2 below, IoT devices can be used to schedule irrigation, manage livestock, and keep an eye on fertilizer and crop health. Farmers can increase productivity, reduce waste, and enhance sustainability by utilizing this cutting-edge IoT technology [50, 51].

The challenges of climate change, food insecurity, and sustainability call for innovative agricultural solutions. For rice crop growth, nutrients like nitrogen are vital for robust leafy development, while phosphorus and potassium support root health and grain quality. Overreliance on generalized fertilizer practices without real-time monitoring can harm soil health, reduce productivity, and cause environmental damage. IoT-based smart farming offers a transformative solution, enabling continuous monitoring of soil parameters through cloud technologies. The IoT approach empowers farmers to make data-driven decisions, optimize fertilizer use, and enhance crop health while minimizing environmental impact [52,53].

2. Methodology & materials

This section outlines the methodology and materials for the proposed IoT-based system aimed at enhancing rice crop management, particularly optimizing fertilizer use and irrigation practices.



Fig. 2. IoT applications in agriculture.

2.1. Proposed methodology and IoT-based system

The proposed system comprises a smart soil sensor, data acquisition system, wireless communication network, and cloud-based platform. The IoT-based system that is proposed in this article includes a smart sensor, data acquisition system, wireless communication network, and a cloud-based platform. This smart soil sensor accurately measures temperature, moisture, conductivity, pH, salinity, and NPK levels. It is built with stainless steel probes, sealed with epoxy resin, and runs on a 5–24 V power supply. The sensor has stable performance and high sensitivity, and can directly and stably reflect the actual moisture content of different soils and the nutritional status of the soil in time, providing a data basis for scientific planning about planting. Data is sent via an Internet connectivity module for efficient real-time monitoring to the cloud platform for analysis and visualization in agriculture. The sensor, equipped with advanced probe technology, simultaneously measures soil temperature, moisture, salinity, EC, pH, nitrogen, phosphorus, and potassium levels. The collected data is transmitted wirelessly. Fig. 3 shows the methodology of the overall system.

Table 3 describes the details of equipment and sensors used in the development of this IoT-based fertilizer and crop management system.

2.2. Experimental setup and soil sampling

A field study was conducted at different locations of rice fields in Rahim Yar Khan, Pakistan to assess soil health and identify potential

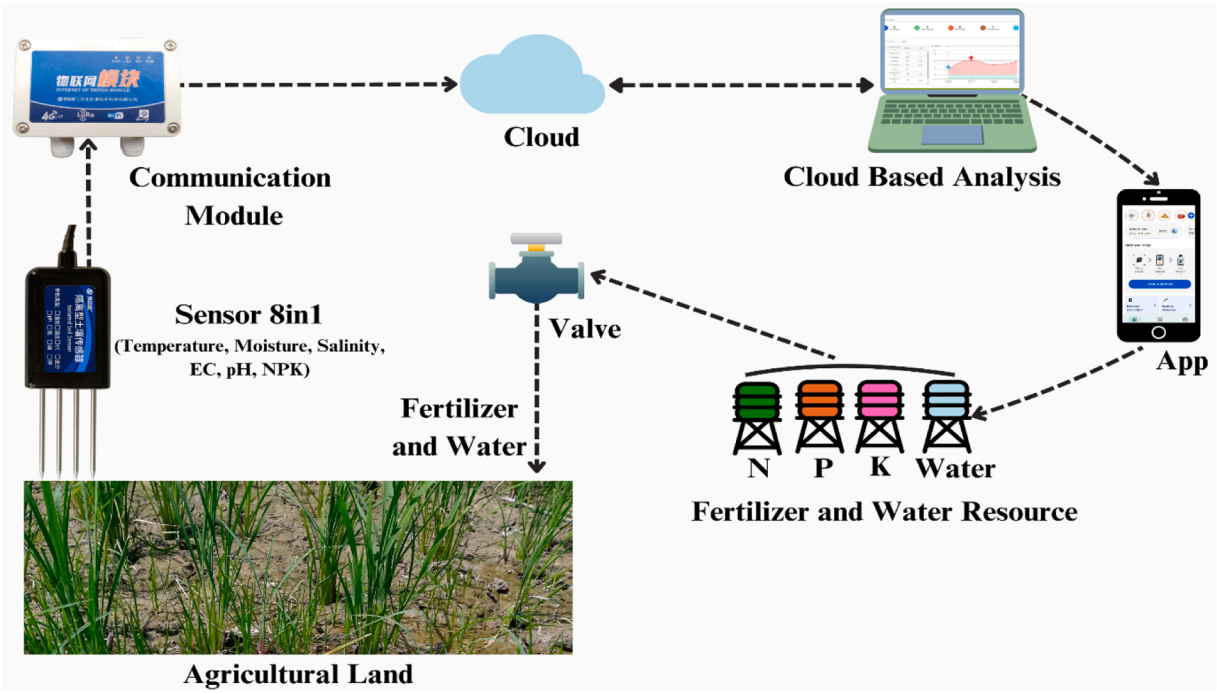


Fig. 3. Proposed framework.

Table 3
Smart soil nutrients monitoring sensor details.

Parameter	Range/Description
Communication Module	Hebei Ousu Electronic Technology Co., Ltd.
Cloud Integration	Jiousu IoT Cloud Server, real-time monitoring, analysis, visualization, CSV data export
IoT Module	ESP8266 kit-based WiFi Enabled Circuit
Sensor Type	8in1 Soil Nutrients Monitoring Sensor
Applications	Agriculture, precision farming, soil monitoring, smart farming, environmental monitoring
Supply Voltage	DC 5–24V
Power Consumption	15 mA @ 12 V DC
Communication Method	RS485
Baud Rate	9600bps
Temperature Range	−40 °C to 85 °C
Temperature Accuracy	±1.0 °C
Soil Moisture Range	0 % to 100 % (Relative Humidity)
Soil Moisture Accuracy	±5 %
Electrical Conductivity (EC) Range	0 to 10,000 µS/cm
EC Accuracy	±5 %
Nutrient Range (NPK)	Nitrogen (N): 0 to 200 mg/kg Potassium (K): 0 to 200 mg/kg Phosphorus (P): 0 to 200 mg/kg
NPK Accuracy	±10 %
pH Range	0 to 14
pH Precision	±10 %
Salt (Salinity) Range	0 to 10,000 µS/cm
Salt Accuracy	±5 %
Sealing Material	Black epoxy resin
Material	Four-pin design with three solid stainless-steel probes, one hollow stainless-steel probe, and zinc alloy. Total length: 138 ± 1 mm; Probe length: 65 ± 1 mm; Diameter: 3 ± 0.2mm

growth constraints using the proposed system. The proposed system was tested through external (laboratory/workshop) and on-site evaluations. Prior to deployment, a site visit was conducted to assess conditions on the farm. The rice crop were in the leaf development stage (end of July to beginning of August). The system was deployed in a 1-acre rice field (28.391257°N, 70.336044°E) under normal weather conditions. A mini solar panel of 20 W with a power bank ensured self-sufficiency, and the internet connection was via a mobile WiFi hotspot. 14 soil samples (0–6 inches deep) were field tested in real time at various locations and selected samples were cross-checked at a local soil testing laboratory for accuracy assessment. The primary objective of the study was to evaluate the effectiveness and accuracy of the proposed IoT based real-time environmental monitoring and data collection system for agricultural applications. Fig. 4 shows the laboratory and field experiment activities, including soil sampling and equipment setup.

3. Experimental results: decision-making with real-time monitoring and AI-driven mobile application

Data on temperature, pH, soil moisture, salinity (µS/cm), EC (µS/cm), nitrogen, phosphorus and potassium levels were recorded for each sample (Table 4). Temperatures ranged from 30.5 °C to 33 °C, reflecting typical seed growing season conditions. Soil moisture content varied between samples, indicating potential water management problems. Salinity was generally moderate, but effects on crop growth require further investigation. Nutritional analysis showed adequate nitrogen and potassium content, while phosphorus content showed some variation. Particularly, soil pH was consistently high in all samples, indicating alkaline conditions and suitable for rice crop growth and productivity.

The performance of the proposed system is further analyzed through a web-based cloud platform that stores data from rice fields, as shown in Fig. 5. Smart soil sensors collect data from soil samples and send them to the online cloud via an Internet connection, where farmers can analyze the data and make smart and effective decisions about irrigation and fertilization activities.

3.1. Algorithm for fertilizer and irrigation recommendation of rice crop

- Function IrrigationAndFertilizerRecommendation(soilMoistureData, nitrogenData, currentDate)
- // Initialize variables
- consecutiveDryDays=0
- consecutiveLowNitrogenDays=0
- irrigationNeeded = False
- fertilizerNeeded = False

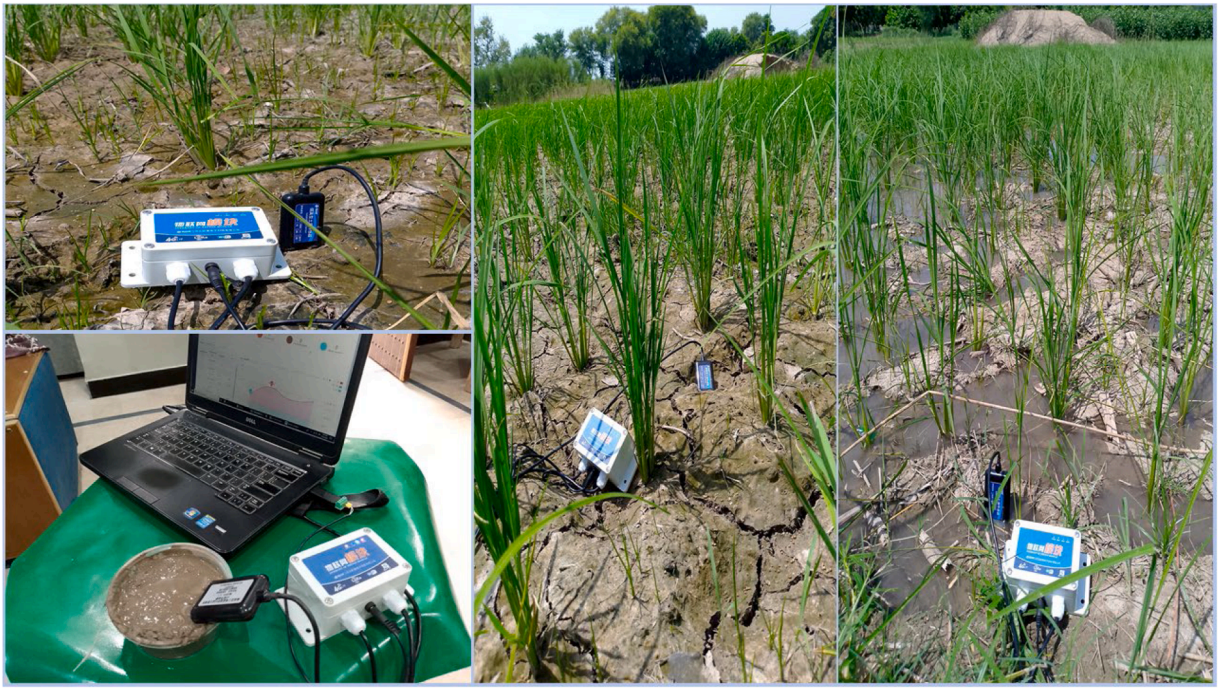


Fig. 4. Field testing in rice crop.

Table 4
Summary of experimental results.

Rice Soil	Temp (°C)	EC (µS/cm)	Soil Moisture (%)	Salinity (µS/cm)	Soil pH	Nitrogen (mg/kg)	Phosphorus (mg/kg)	Potassium (mg/kg)
Soil Sample 1	31.1	1884	60.6	1712	7.87	71	18	101
Soil Sample 2	33.2	1852	65.2	1701	7.13	83	16	119
Soil Sample 3	30.5	1864	62.3	1718	8.17	103	15	116
Soil Sample 4	30.8	1672	67.2	1217	7.18	94	16	109
Soil Sample 5	31.8	1271	81.2	1244	8.11	78	15	117
Soil Sample 6	32.1	1184	90.6	1212	7.81	98	18	141
Soil Sample 7	32.2	1191	94.1	1231	8.05	101	18	126
Soil Sample 8	32.9	1246	80.0	1388	7.19	77	19	124
Soil Sample 9	32.8	1252	87.2	1301	8.33	81	16	139
Soil Sample 10	31.5	1211	81.8	1362	8.16	84	15	121
Soil Sample 11	31.7	1316	80.0	1488	7.79	77	19	125
Soil Sample 12	32.9	1276	82.0	1388	7.52	75	18	123
Soil Sample 13	31.1	1724	83.2	1733	7.91	88	16	131
Soil Sample 14	32	1733	79.4	1714	8.21	83	19	136

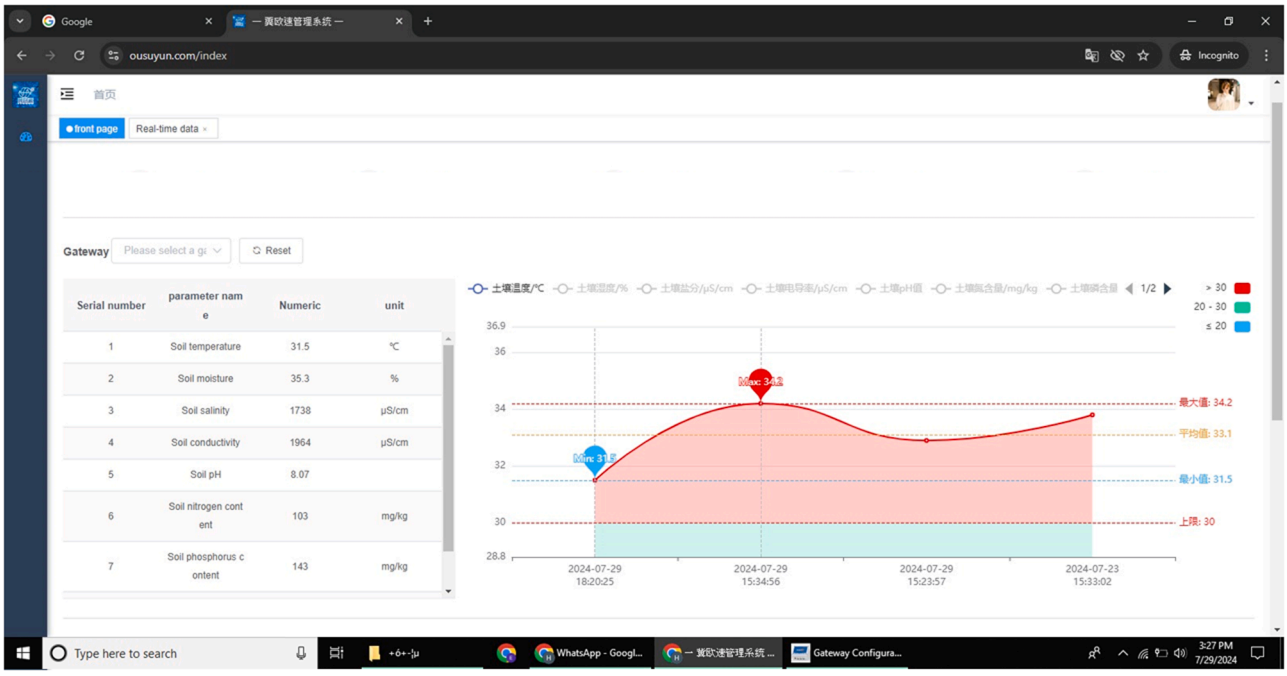


Fig. 5. Cloud platform for real-time monitoring.

- fieldCapacityThreshold = 70 // Optimal soil moisture for rice
- nitrogenDropThreshold = 100 // Significant drop in nitrogen levels

// Check soil moisture and nitrogen levels for the past 7 days

- for *i* = currentDate - 6 to currentDate
- if soilMoistureData[i] < fieldCapacityThreshold
- consecutiveDryDays=consecutiveDryDays /1
- else
- consecutiveDryDays=0
- if nitrogenData[i] < nitrogenData[i-1] - nitrogenDropThreshold
- consecutiveLowNitrogenDays=consecutiveLowNitrogenDays+ 1
- else
- consecutiveLowNitrogenDays=0

// Check if irrigation is needed

- if consecutiveDryDays≥7 // Trigger irrigation after 7 consecutive dry days
- irrigationNeeded = True
- CalculateIrrigationAmount(soilType, cropStage, weatherConditions)
- ApplyIrrigation()

// Check if fertilizer is needed

- if consecutiveLowNitrogenDays≥ 7 // Trigger fertilizer after 7 consecutive low nitrogen days
- fertilizerNeeded = True
- RecommendNitrogenFertilizer(cropStage, soilConditions)
- ApplyFertilizer()
- End Function

Explanation of Algorithm: The function observes soil moisture and nitrogen contents for 7 days. On consecutively dry days exceeding 7 or nitrogen contents dropping too low, it invokes irrigation or fertilizer applications. This IoT-based data-driven approach confirms rice crops obtain timely care for optimal growth and sustainable production. Mobile applications and software can be developed considering this idea and algorithm.

3.2. Plantix: the AI-Driven plant doctor app

Plantix is like having a crop expert in your pocket. This smart app uses AI to diagnose plant health issues, recommend fertilizers, and offer pest control tips [54,55]. Analyzing soil and environmental data, gives farmers easy-to-understand, tailored advice to help improve their crops and boost sustainability. To calculate the required amount of fertilizer and rice plant disease diagnosis using Plantix mobile app, data from the proposed system was entered into this mobile application Plantix. This application use soil test data to determine the precise fertilizer needs for rice crops. Figs. 6 and 7 illustrate snapshots of this application, showing recommended fertilizer rates and diagnosis based on the soil data analyzed.

4. Findings and discussion of essential requirements

The results of soil analysis of rice crops reveal several important insights regarding soil temperature, pH, moisture content, salinity, EC and nutrient levels. The data offers insights into the variability of these parameters across different soil samples, which are crucial for understanding soil health and optimizing agricultural practices. These factors jointly affect crop metabolism, growth, and productivity. The most important parameters for rice crops are discussed below.

Soil Temperature: The soil temperature across the samples ranged from 30.5 °C to 33.2 °C, with an average of approximately 31.8 °C. This range is typical for rice paddies, where higher temperatures can influence microbial activity and nutrient availability. Temperature is an important factor affecting the growth of rice, and the measured temperatures suggest a generally favorable condition for rice cultivation.

Soil Moisture: Soil moisture values varied from 60.6 % to 94.1 %, showing a wide range of water retention across samples. Sample 7 exhibited the highest moisture content at 94.1 %, which indicates well-watered conditions, potentially beneficial for rice cultivation. However, excess moisture can also lead to waterlogging, which may affect root respiration and nutrient uptake. Conversely, sample 1 had a lower moisture content (60.6 %), which may indicate dryer conditions, requiring irrigation management for optimal rice growth.

Soil pH: The soil pH ranged from 7.13 to 8.33, indicating a generally neutral to slightly alkaline pH across the samples. Rice plants typically thrive in slightly acidic to neutral soil (pH 5.5–7.0), but the observed pH values are within acceptable limits for rice cultivation. Samples with pH values above 8 (e.g. 3, 6, 7, and 13) indicate a risk of reduced absorption of nutrients, especially phosphorus, because high alkalinity may hinder its availability, potentially affecting crop metabolism and growth.

Nutrient Content:

- **Nitrogen:** The nitrogen content ranged from 71 mg/kg to 103 mg/kg. Nitrogen is a vital nutrient for rice growth, and the measured levels are adequate for promoting vegetative growth. Sample 3 had the highest nitrogen content at 103 mg/kg, which could support strong vegetative development.
- **Phosphorus:** Phosphorus values were relatively uniform, ranging from 15 mg/kg to 19 mg/kg, with sample 14 having the highest phosphorus content at 19 mg/kg. Phosphorus is essential for root development and flowering in rice, and the measured values are within a typical range for healthy rice soils.
- **Potassium:** Potassium levels ranged from 101 mg/kg to 141 mg/kg, with sample 6 having the highest value of 141 mg/kg. Potassium is crucial for regulating plant water balance and improving disease resistance, and the observed levels are sufficient for supporting optimal growth and yield.

The variation in these parameters across the different soil samples underscores the need for tailored management practices in rice farming. While some samples exhibit favorable conditions for rice growth, others may require adjustments in irrigation, fertilization, or pH management to optimize productivity. For instance, soil samples with higher EC and salinity may need strategies to manage soil salinity, while those with lower moisture content might require additional irrigation. The dataset highlights the complex interplay of environmental factors and nutrients in determining the sustainability and productivity of rice farming. The 8-in-1 soil sensor dataset provides valuable insights into the soil conditions and nutrient availability for sustainable rice farming. These parameters

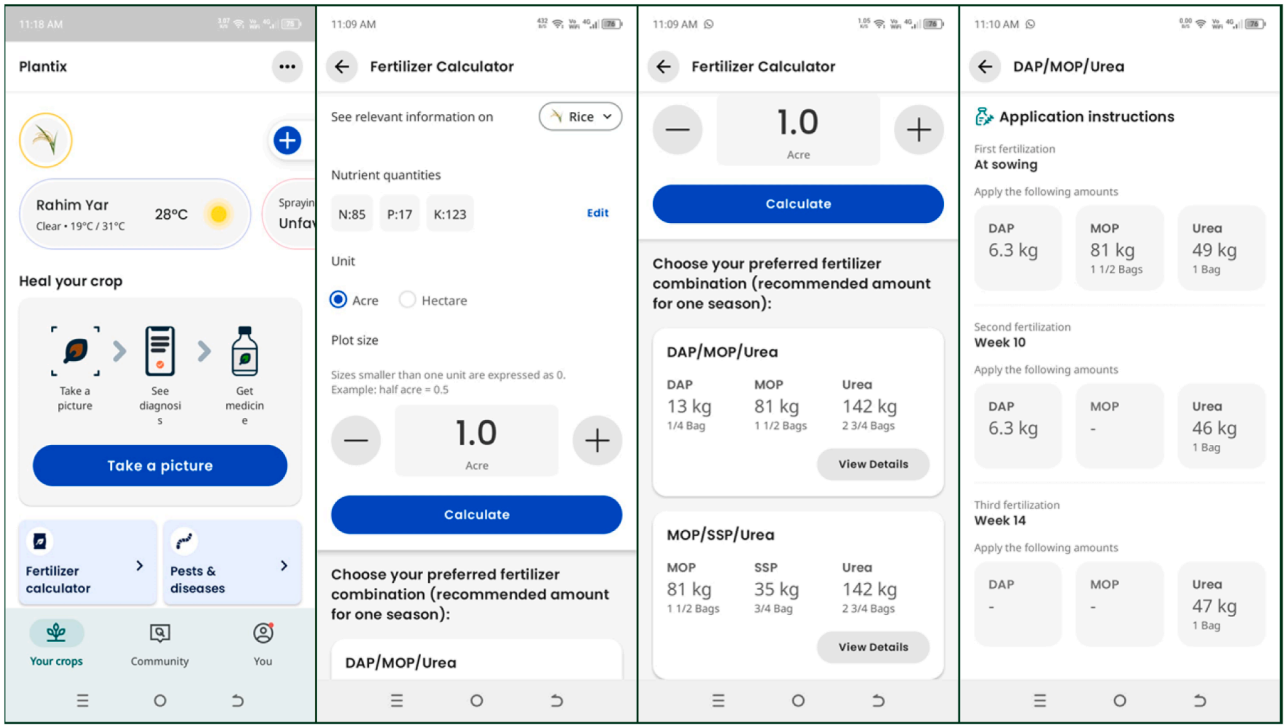


Fig. 6. Plantix mobile app.

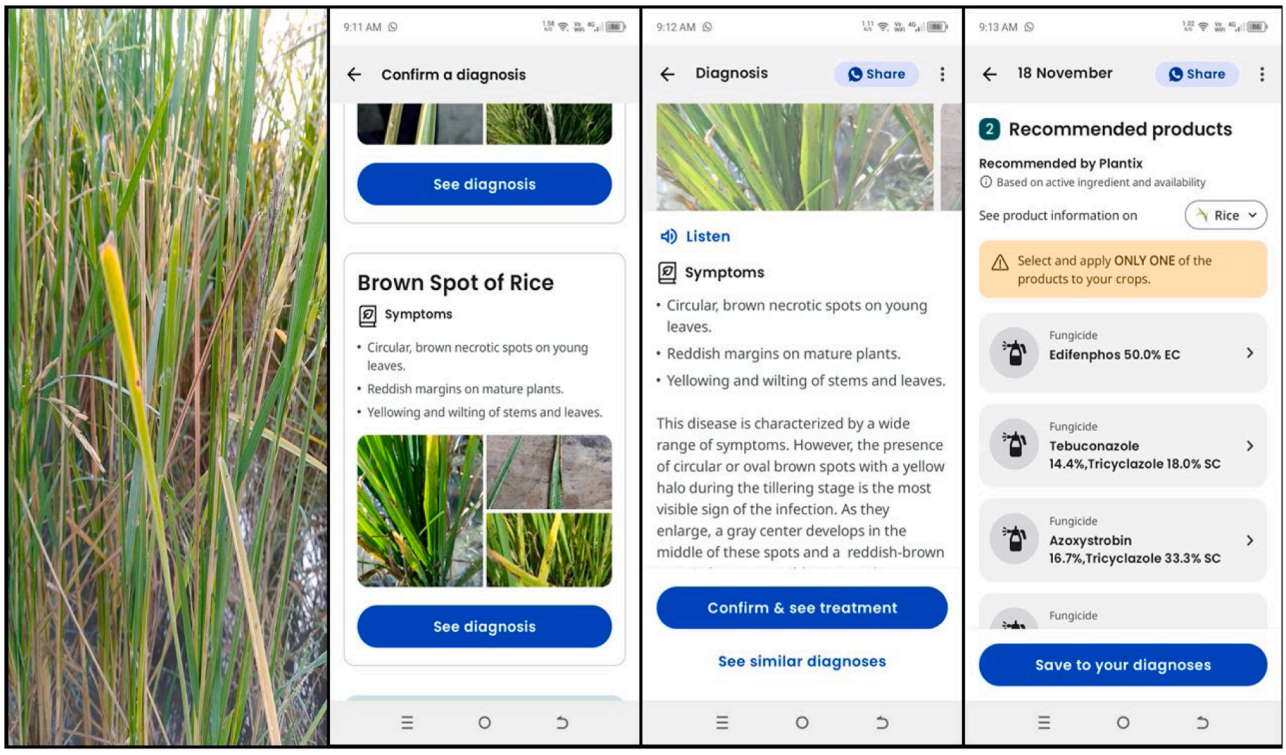


Fig. 7. Rice plants disease diagnosis using Plantix mobile app.

can guide informed decision-making for improving crop management practices, enhancing productivity, and ensuring environmental sustainability in rice cultivation.

5. Conclusions

IoT-based smart farming research presented in this article integrates metabolic monitoring of rice plants with key soil parameters: temperature, moisture, salinity, electrical conductivity (EC), pH, nitrogen, phosphorus, and potassium. By providing farmers with actionable insights, this system empowers them to make data-driven decisions, thereby increasing crop growth, development and grain yield. Field tests on rice crops in Rahim Yar Khan, Pakistan, validate the system's effectiveness, showing how real-time data monitoring enables precise adjustments in crop management to optimize growing conditions. The variation in soil parameters observed during testing, such as moisture levels and pH, shows the importance of continuous monitoring to maintain optimal conditions for rice crop metabolism and productivity. By integrating IoT with cloud computing, predictive algorithm and AI-driven mobile application, the system provides a scalable solution for modern agriculture, enabling efficient use of resources, reduced environmental impact, and improved crop resilience. This research addresses the need for data-driven decision-making in agriculture, paving the way for more sustainable and productive agricultural practices that can meet the demands of the world's growing population.

Ethical statement

The research described in this article was performed and conducted ethically, adhering to relevant guidelines and ensuring no harm to humans, animals, or the environment.

CRediT authorship contribution statement

Hammad Shahab: Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Muhammad Naeem:** Software, Methodology, Formal analysis. **Muhammad Iqbal:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition. **Muhammad Aqeel:** Writing – original draft, Software, Methodology, Investigation, Data curation. **Syed Sajid Ullah:** Resources, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare no conflicts of interest affecting the integrity of this work.

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Data availability

No data was used for the research described in the article.

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